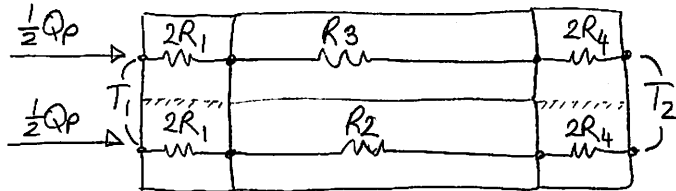


① Parallel Approach

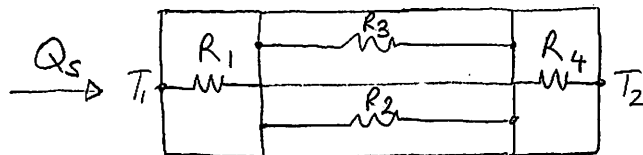


$$\left. \begin{aligned} R_L &= \frac{L_1}{k_1 A_2} + \frac{L_2}{k_2 A_2} + \frac{L_4}{k_4 A_2} \\ R_u &= \frac{L_1}{k_1 A_3} + \frac{L_2}{k_3 A_3} + \frac{L_4}{k_4 A_3} \end{aligned} \right\} \frac{1}{R} = \frac{1}{R_L} + \frac{1}{R_u}$$

$$\frac{1}{R_p} = \left(\frac{A_2}{L_1/k_1 + L_2/k_2 + L_4/k_4} + \frac{A_3}{L_1/k_1 + L_3/k_3 + L_4/k_4} \right)$$

$$Q_p = \frac{\Delta T}{R_p} = \left(\frac{A_2}{L_1/k_1 + L_2/k_2 + L_4/k_4} + \frac{A_3}{L_1/k_1 + L_3/k_3 + L_4/k_4} \right) \Delta T$$

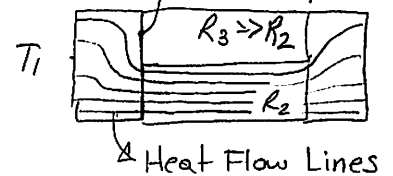
② Series Approach



$$R_s = \frac{L_1}{k_1 A_1} + \left(\frac{k_2 A_2}{L_2} + \frac{k_3 A_3}{L_2} \right)^{-1} + \frac{L_4}{k_4 A_1}$$

$$Q_s = \frac{\Delta T}{R_s} = \left(\frac{L_1}{k_1 A_1} + \frac{L_2}{k_2 A_2 + k_3 A_3} + \frac{L_4}{k_4 A_1} \right)^{-1} \Delta T$$

Aside: Think of this way:
Real Life: T will vary here



It is always true that:

$$Q_p \leq Q_{\text{REAL}} \leq Q_s$$

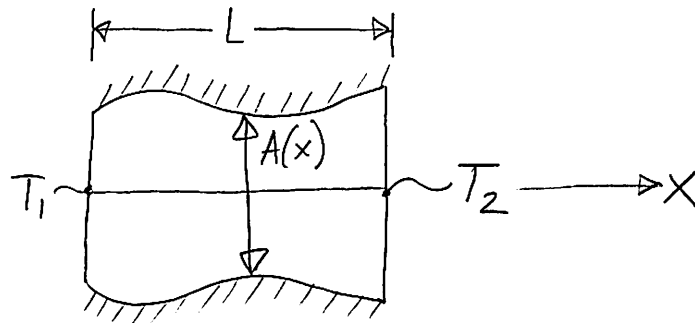
↳ Lower Bound estimate

↳ Upper Bound estimate.

In general:

- 1) Adding adiabatic surfaces: $R \uparrow$, $Q \downarrow$
- 2) Allowing infinite conductance: $R \downarrow$, $Q \uparrow$

Example #6



$$dR = \frac{dx}{kA(x)} \rightarrow R \geq \frac{1}{k} \int_0^L \frac{dx}{A(x)}$$

$$Q \leq \frac{\Delta T}{\frac{1}{k} \int_0^L \frac{dx}{A(x)}}$$

Assuming it's a 1D problem.
In reality, 2D and added resistance will come into play due to longer path.

The Nievé approach: $\bar{A} = \frac{1}{L} \int_0^L A(x) dx \rightarrow R \approx \frac{L^2}{k \int_0^L A(x) dx}$

$$Q \approx \frac{k \Delta T}{L^2} \int_0^L A(x) dx$$

But, it's important to note that:

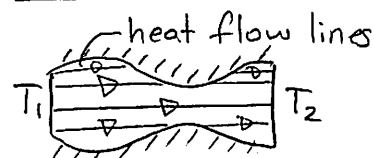
$$\underbrace{\frac{1}{L} \int_0^L A(x) dx}_{\text{Arithmetic Mean}} \geq \underbrace{\left(\frac{1}{L} \int_0^L \frac{dx}{A(x)} \right)^{-1}}_{\text{Harmonic Mean}}$$

$$R_{\text{real}} \geq \frac{1}{k} \int_0^L \frac{dx}{A(x)} \geq \frac{L}{k} \left(\frac{1}{L} \int_0^L A(x) dx \right)^{-1}$$

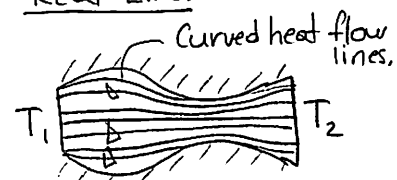
$$Q_{\text{real}} \leq \frac{\Delta T}{\frac{1}{k} \int_0^L \frac{dx}{A(x)}} \leq \frac{k \Delta T}{L^2} \int_0^L A(x) dx$$

Aside:

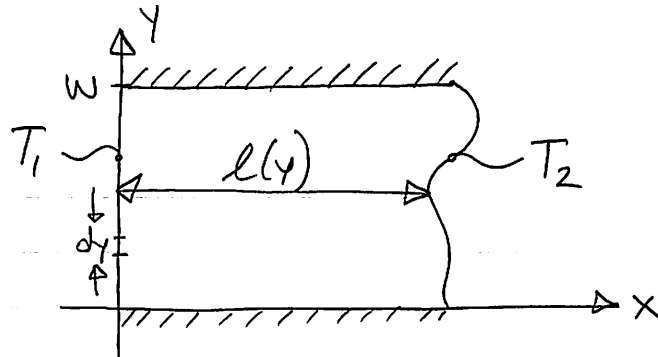
1-D solution



Real Life:



Example #7



Q' = heat transfer rate per unit length into board or paper.

$$dQ' \approx k dy \frac{\Delta T}{l(y)}$$

$$Q' \geq k \Delta T \int_0^w \frac{dy}{l(y)}$$

$$R' \leq \frac{1}{k} \left(\int_0^w \frac{dy}{l(y)} \right)^{-1}$$

The Nieve' Approach:

$$\bar{l} \approx \frac{1}{w} \int_0^w l(y) dy$$

$$\bar{R} \approx \frac{1}{kw} \int_0^w \frac{dy}{l(y)}$$

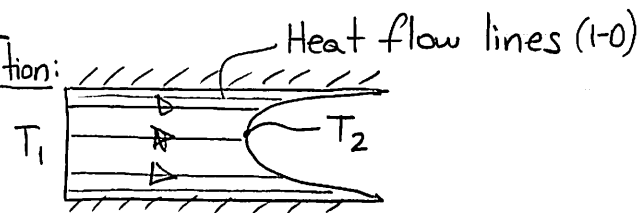
$$Q' \approx kw \Delta T \left(\int_0^w \frac{dy}{l(y)} \right)^{-1}$$

Real heat transfer.

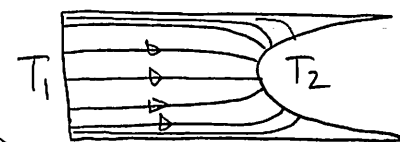
$Q' \geq (\dots)$ since in reality, heat can spread laterally and take the path of least resistance. This is not considered here.

i.e.

Solution:



Real Life:



From this we can say that:

$$R'_{\text{real}} \leq \frac{1}{k} \left(\int_0^w \frac{dy}{l(y)} \right)^{-1} \leq \frac{1}{kw} \int_0^w \frac{dy}{l(y)}$$

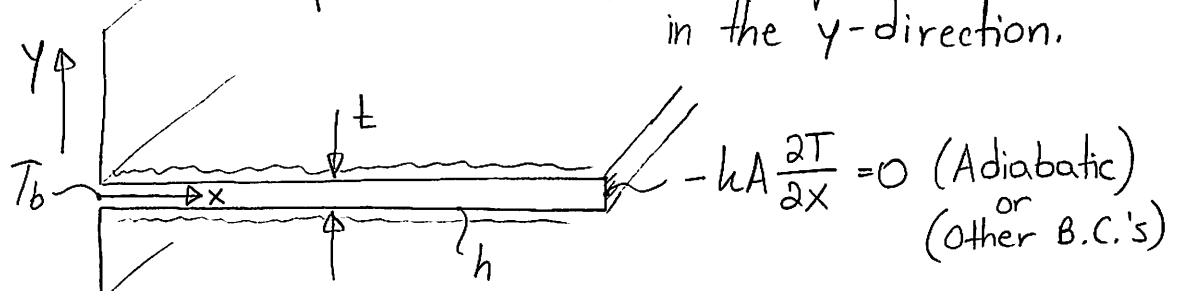
$$Q'_{\text{real}} \geq k \Delta T \int_0^w \frac{dy}{l(y)} \geq kw \Delta T \left(\int_0^w \frac{dy}{l(y)} \right)^{-1}$$

$\Rightarrow$ We've already learned that Arithmetic mean $\geq$ Harmonic mean.

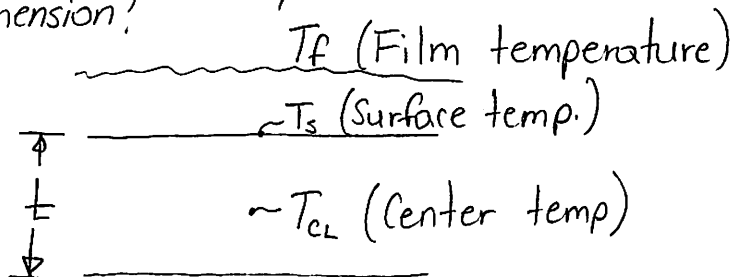
Quasi - 1D Conduction : Fins

Assumptions; 1) $\rho, A = \text{constant}$ (Perimeter and Area of Fin)
 2) No heat generation
 3) The heat transfer is 1-D

What does assumption 3) mean? No temperature variation in the y -direction.



So how do we quantify the temperature variation in the y -dimension?



Doing a heat balance across the fin, we obtain:

$$\underbrace{-k \frac{\partial T}{\partial y} (L dx)}_{\text{Conduction}} = \underbrace{h (L dx) (T_s - T_f)}_{\text{Convection}}$$

$$\frac{\partial T}{\partial y} = \frac{T_{CL} - T_s}{t/2} = h (T_s - T_f)$$

$$\frac{T_{CL} - T_s}{T_s - T_f} = -\frac{1}{2} \frac{h t}{k} \approx 0.05$$

$$\frac{1}{2} \left(\frac{h t}{k} \right) \leq \frac{1}{20} \Rightarrow \boxed{\frac{h t}{k} \leq \frac{1}{10} = Bi_t} \Rightarrow \text{for fin heat transfer in the thickness direct. Biot Number. } (48)$$

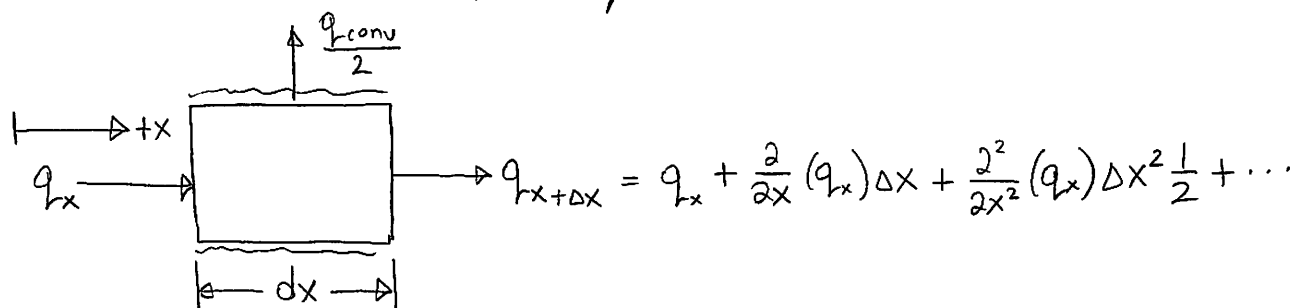
Interpretation: 1) Temp. changes across the fin thickness are small compared to those external to the fin.
(isothermal in y , at a particular x)
2) Internal resistance to conduction across the thickness of the fin is small compared to resistance due to external convection. i.e. $R_{cond} \ll R_{conv}$

We can double check this right away:

$$\frac{R_{cond}}{R_{conv}} = \frac{\frac{t}{kA}}{\frac{1}{hA}} = \frac{ht}{k} \ll 1$$

$\underbrace{\hspace{1cm}}_{Bi_t} \ll 1$

Now we can model the temperature distribution:



$$q_{conv} = hPdx(T - T_f) \quad ; \quad P = \text{perimeter or surface area/unit length}$$

We also know that: $q_x = -kA \frac{\partial T}{\partial x}$

Energy balance:

$$q_x - \left\{ q_x + \frac{\partial q_x}{\partial x} \Delta x + \frac{1}{2} \frac{\partial^2 q_x}{\partial x^2} \Delta x^2 + \dots \right\} = hP\Delta x(T - T_f)$$

$$-\frac{\partial q_x}{\partial x} \cdot dx = hPdx(T - T_f)$$

$$\frac{d}{dx} \left(kA \frac{\partial T}{\partial x} \right) - hP(T - T_f) = 0$$

Defining $\theta = T - T_f$, $d\theta = dT$

$$kA \frac{\partial^2 \theta}{\partial x^2} - hP\theta = 0 \Rightarrow \frac{\partial^2 \theta}{\partial x^2} - \frac{hP}{kA} \theta = 0$$

$$\frac{\partial^2 \theta}{\partial x^2} - m^2 \theta = 0$$

$$\text{Let } \boxed{\frac{hP}{kA} = m^2}$$

Works if:

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = 0$$

Linear, second order ODE

Our characteristic equation is: (ODE concept) Euler did it!

$$\lambda^2 - m^2 = 0$$

$$\lambda^2 = m^2$$

$$\lambda = \pm m$$

So our general solution is:

$$\theta(x) = C_1 e^{mx} + C_2 e^{-mx}$$

Aside: Proof of solution:

$$\theta = C_1 e^{mx} + C_2 e^{-mx}$$

$$\theta' = mC_1 e^{mx} - mC_2 e^{-mx}$$

$$\theta'' = m^2 C_1 e^{mx} + m^2 C_2 e^{-mx}$$

$$\theta'' - m^2 \theta = m^2 (C_1 e^{mx} + C_2 e^{-mx}) - m^2 \theta = 0 \quad \text{QED}$$

We have four directions to go to:

① Infinite Fin case ($x \rightarrow \infty$)

Our B.C.'s become:

$$\theta(x=0) = T_b - T_f = \theta_b$$

$$\theta(x=L \rightarrow \infty) = T_f - T_f = 0$$

$$\theta(x) = C_1 e^{mx} + C_2 e^{-mx}$$

$$\theta(\infty) = 0 = \underbrace{C_1 e^{m(\infty)}}_{C_1=0} + C_2 e^{-m(\infty)}$$

$$\theta(x=0) = \theta_b \quad C_1 = 0 \text{ for solution to be valid}$$

$$\theta_b = T_b - T_f = C_2 e^{-m(0)} \Rightarrow C_2 = \theta_b$$

$$\theta(x) = \theta_b e^{-mx}$$

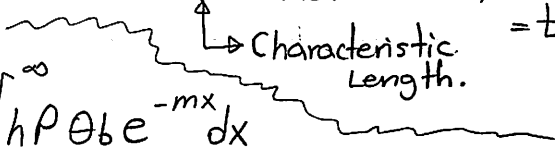
$$\boxed{\frac{T - T_f}{T_b - T_f} = e^{-\frac{hP}{kA} x}}$$

What about heat transfer from the base to the fin?

$$Q_{\text{net}} = Q(0) = -kA \left. \frac{\partial \theta}{\partial x} \right|_{x=0} = +kA \theta_b m e^{-m(0)}$$

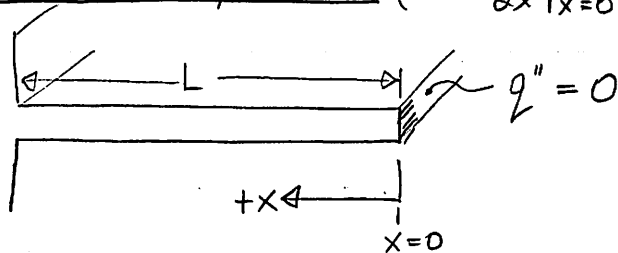
$$Q_{net} = +kA\theta_b \sqrt{\frac{hP'}{kA}} = \sqrt{kAhP'}\theta_b$$

Note: $\frac{Q_{net}}{Q_o} = \frac{\sqrt{kAhP'}\theta_b}{hA\theta_b} = \sqrt{\frac{kP'}{hA}} = \sqrt{\frac{k}{ht}} = Bi_t^{-1/2} \gg 1$

Also note here that: $\frac{P}{A} = \frac{1}{t}$. In general $L_c = \frac{V_{body}}{A_{surface}} = \frac{AK}{PK} = \frac{A}{P} = t$


Sanity check: $Q_{net} = \int_0^\infty hP\theta dx = \int_0^\infty hP\theta_b e^{-mx} dx$
 $= hP\theta_b \frac{e^{-mx}}{m} \Big|_0^\infty = \frac{hP\theta_b}{m} = \sqrt{kAhP'}\theta_b$
 $\Downarrow$
 Works!

② Insulated Tip case $(-k \frac{\partial T}{\partial x} \Big|_{x=0} = 0)$



B.C.'s:
 $-k \frac{\partial \theta}{\partial x} \Big|_{x=0} = 0$
 $\theta(x=L) = \theta_b$

$$\theta(x) = C_1 e^{mx} + C_2 e^{-mx}$$

We can rewrite this as:

$$\theta(x) = C_1 \cosh(mx) + C_2 \sinh(mx)$$

Substitute in our B.C.'s

$$\frac{\partial \theta}{\partial x} = C_1 m \sinh(mx) + C_2 m \cosh(mx)$$

$$\frac{\partial \theta}{\partial x} \Big|_{x=0} = C_1 m \sinh(0) + C_2 m \cosh(0) \Rightarrow C_2 m = 0 \Rightarrow C_2 = 0$$

$$\theta(x) = C_1 \cosh(mx) \Rightarrow \text{Apply our second B.C.}$$

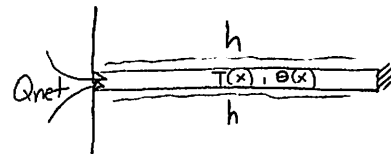
$$\theta(x=L) = \theta_b = C_1 \cosh(mL)$$

$$C_1 = \frac{\theta_b}{\cosh(mL)}$$

Back substituting

$$\Theta(x) = \frac{\Theta_b \cosh(mx)}{\cosh(mL)}$$

$$\boxed{\frac{T - T_f}{T_b - T_f} = \frac{\cosh(mx)}{\cosh(mL)}}$$



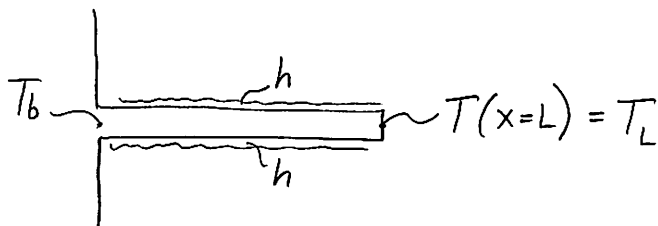
How about heat transfer?

$$Q_{net} = -kA \left. \frac{\partial \Theta}{\partial x} \right|_{x=L} = +kA \Theta_b m \frac{\sinh(mL)}{\cosh(mL)} = kA \Theta_b \sqrt{\frac{hP}{kA}} \frac{\sinh(mL)}{\cosh(mL)}$$

$$\boxed{Q_{net} = \sqrt{kAhP} (T_b - T_f) \frac{\sinh(mL)}{\cosh(mL)}} \Rightarrow \text{Heat transfer from the insulated tip fin.}$$

$\tanh(mL)$

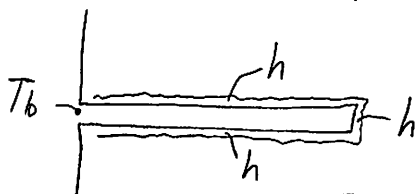
③ Prescribed Tip Temperature ($T(x=L) = T_L$)



I won't go through the math, (you can do this) but the solution is:

$$\boxed{\frac{T - T_f}{T_b - T_f} = \frac{T_L - T_f}{T_b - T_f} \frac{\sinh(mx) + \sinh[m(L-x)]}{\sinh(mL)}}$$

④ Convection on Tip ($h(T(x=L) - T_f) = -k \left. \frac{\partial T}{\partial x} \right|_{x=L}$)



$$\boxed{\frac{T - T_f}{T_b - T_f} = \frac{\cosh(m(L-x)) + \frac{h}{mk} \sinh(m(L-x))}{\cosh(mL) + \frac{h}{mk} \sinh(mL)}}$$

Fin Efficiency (Insulated tip)

$$\text{Fin Efficiency} = \eta_{fin} = \frac{\text{Actual Heat Transfer}}{\text{Heat Transfer if } \theta = \theta_b \text{ everywhere}}$$

$$= \frac{q_{actual}}{q_{ideal}}$$

We know: $q_{ideal} = h(PL)(T_b - T_f)$

$$q_{actual} = \int_0^L hP(T - T_f)dx = \underbrace{\sqrt{kAPh} \theta_b \tanh(mL)}_{\text{Insulated Tip } Q_{net}}$$

$$\eta_{fin} = \frac{\sqrt{kAPh} \theta_b \tanh(mL)}{hPL \theta_b} = \underbrace{\sqrt{\frac{kA}{hP}}}_{\frac{1}{m}} \cdot \frac{\tanh(mL)}{L}$$

$$\boxed{\eta_{fin} = \frac{\tanh(mL)}{mL}} = \frac{q_{actual}}{q_{ideal}}$$

For other configurations, see Mills pg. 104-105

Fin Resistance (R_{fin}) $\Rightarrow$ For insulated tip case

$$R_{fin} = \frac{\theta_b}{q_{actual}} = \frac{T_b - T_f}{q_{actual}} = \frac{\theta_b}{\sqrt{kAPh} \tanh(mL) \theta_b} = \frac{1}{h A_{fin} \eta_{fin}}$$

$$\boxed{R_{fin} = \frac{1}{h A_{fin} \eta_{fin}}}$$

A_{fin} = outside area of fin (PL)
 η_{fin} = fin efficiency
 h = heat transfer coeff.

Fin Effectiveness (ϵ_{fin})

$$\epsilon_f = \frac{\text{actual heat transfer}}{\text{heat transfer if no fin}}$$

$$= \frac{q_{ideal} \cdot \eta_f}{q_{ideal} \left(\frac{A}{PL} \right)} = \eta_f \left(\frac{PL}{A} \right) \Rightarrow \boxed{\epsilon_f = \eta_f \frac{PL}{A}}$$

P = perimeter
 L = fin length
 A = c.s.A. of fin.